Lilith M. Holland

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Education

University of West Florida — Pensacola, FL
Master of Science in Data Science — *2024 – 2025*

University of South Florida — Tampa, FL
Bachelor of Science in Information Science — *2021 – 2023*

Experience

Associate Data Scientist — June 2023 – Sept. 2024
Luminosity Data Analytics — *St. Petersburg, FL*

- Targeted SQL data mart for pilot A/B testing, to support Difference-in-Differences and uplift modeling.
- Technical lifecycle management for 3 clients, transforming unstructured data to production inputs.
- Relational SQL migration of 30 years of historical records parsed from 15,000+ PDF pages.
- Business logic refinement and QA validation across 7 public and internal Power BI dashboards.

Freelance Data Scientist — July 2022 – May 2023
Journal of the International Ombuds Association

- End-to-end NLP pipeline to ingest, clean, and format 400 unstructured academic journals.
- Word2Vec model training using Skip-gram to discover latent word relationships and contextual expansion.
- LDA and cosine similarity algorithms implemented to programatically cluster papers and map thematic gaps.

Student Research Assistant — June 2022 – May 2023
Big Data Analytics & Web Intelligence Research Lab — *Tampa, FL*

- Scalable big data pipeline to ingest, parse, and evaluate 30 Terabytes of nested Twitter JSON data.
- Recursive Python JSON parser developed to handle variable payloads and maintain schema alignment.
- Predictive machine learning models tuned for bot classification and graph network mapping to isolate key actors.

Projects

GapMap | *Python, PyTorch, Transformers, PyMuPDF, Pydantic*

- Document intelligence pipeline designed to ingest PDFs, extract structural layouts, and perform NLP tasks.
- Structural layout extraction using PyMuPDF and PP-DocLayoutV3 to capture reading order and bounding boxes.
- Thematic information retrieval deploying SciBERT for domain-specific embeddings and anchor-word expansion.
- Automated document clustering integrating BERTopic to identify latent corpus themes and map research gaps.

GNN Predictive Weather Modeling | *Python, PyTorch, Matplotlib, Numpy*

- Spatiotemporal Graph Neural Network designed and trained to investigate weather station forecasting capabilities.
- Next-step temperature evaluation using real-world geospatial data from a southwestern Kansas weather network.
- Neural network performance benchmarking against a univariate time-series baseline to isolate accuracy gains.

Directed LLM Graph | *Python, FastAPI, Neo4j, Minio, vLLM, TEI*

- Microservice-based framework designed using Docker Compose to deploy a scalable, local LLM pipeline.
- vLLM and TEI integration via FastAPI to serve low-latency open-source models and embeddings.
- Hybrid Graph RAG storage utilizing Neo4j for network relationship mapping and MinIO for object storage.

Technical Skills

Languages: Python, SQL (MS SQL Server), R
Developer Tools: Git, Docker, Power BI, PyCharm, AWS, Neo4j
Libraries: Pandas, Scikit-Learn, TensorFlow/Keras, PyTorch, Polars, NumPy, Matplotlib, SciPy, FastAPI
Machine Learning: ETL/ELT Pipelines, Llama.cpp, TEI